Brand Bucket Command Center

Master Architecture, Operational Engines & Specifications Manual

Version 2.0.0 (Production Enterprise Edition) • Primary Territory: Nepal

NEXT.JS 14 APP ROUTER | TYPESCRIPT | BIKRAM SAMBAT BS 2083 | SUPABASE DB | CATALOG-DRIVEN EXTRACTOR | SMART SELLING ASSIST | KARMA SCORE 0-100

Brand Bucket Command Center — Master System & Operations Reference Manual

> **Comprehensive Technical Architecture & Operations Guide**

> *Platform: Brand Bucket Command Center 2.0 (Production Enterprise Edition)*

> *Technology Stack: Next.js 14 (App Router), TypeScript, Tailwind CSS, Supabase / Modular DB Service, IndexedDB (PWA Offline Support)*

> *Primary Market: Nepal (Kathmandu Valley & Outstations)*

> *Calendar Systems: Dual Bikram Sambat (BS 2083) & Gregorian (AD)*

Table of Contents

1. Executive Summary & System Architecture
2. Dual Calendar System (Bikram Sambat & Gregorian AD)
3. Role-Based Access Control & User Roles
4. Dynamic WhatsApp Extractor Engine (Fix 67 & 72)
5. Daily Orders Live Cross-Check Engine (Fix 68)
6. Smart Selling Assist Engine (Fix 46)
7. Customer Karma & 360 Intelligence (Fix 69)
8. Offline Field Sales & Beat Operations (Fix 44, 45, 48, 49)
9. Store Health Score Engine (Fix 34)
0. Marketing & Influencer Seeding Pipeline (Fix 60 & 71)
1. Product, Combo & Pricing Catalog Management (Fix 70b & 72)

2. Order Delivery Failures vs. True Returns (Fix 22 & 23)
3. Operations Core: Tasks, Calendar & Audit Activity (Fix 37, 38, 66)
4. Automated Cron Services & Holiday Calendar (Fix 32 & 33)
5. Excel Data Import & Master Google Sheets Export (Fix 39, 42, 43)
6. Production Database Schema & Performance Indexing (Fix 59)
7. Complete Route Map & File Directory Index

---

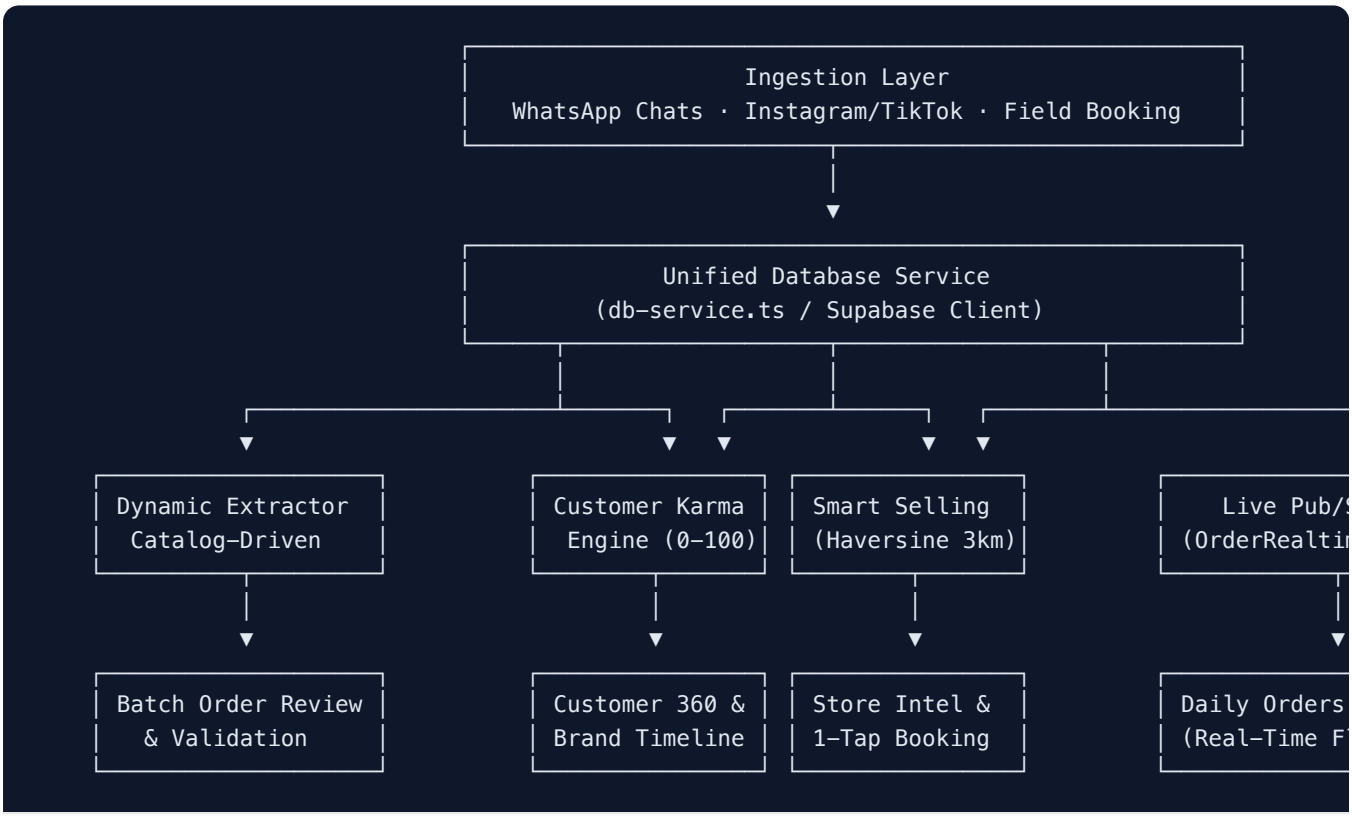
1. Executive Summary & System Architecture

The **Brand Bucket Command Center** is a multi-brand distribution and omnichannel management platform engineered specifically for consumer FMCG and cosmetics brands operating in Nepal (including **Forya**, **Pee Safe & FURR**, and **Renée Cosmetics**).

The platform unifies two traditionally disconnected operational worlds into a single real-time dashboard:

1. **D2C & Social Commerce**: High-volume retail orders originating from WhatsApp, Instagram DMs, Facebook Messenger, TikTok, and Web.
2. **Offline Retail & General Trade (GT / MT)**: Field sales representatives visiting retail pharmacies, modern supermarkets (Bhatbhateni, Saleways), and regional distributor stockists across Kathmandu Valley and outstation provinces.

High-Level System Architecture



---

2. Dual Calendar System (Bikram Sambat & Gregorian AD)

- **Source Code**: `src/lib/nepali-calendar.ts`

In Nepal, government compliance, tax filing, distributor settlements, and consumer holiday shopping follow the **Bikram Sambat (BS)** solar calendar (approx. 56.7 years ahead of Gregorian AD). The platform provides bidirectional conversions without external dependencies:

Mathematical Conversion Model

- **Reference Anchor Point**: `2080-01-01 BS` \leftrightarrow `2023-04-14 AD`.
- **Month Durations**: Nepali months (Baisakh, Jestha, Ashadh, Shrawan, Bhadra, Ashwin, Kartik, Mangsir, Poush, Magh, Falgun, Chaitra) have dynamic lengths ranging from 29 to 32 days depending on the astronomical solar transit.
- **Conversion Methods**:
- `adToBs(dateAd: Date | string)`: Computes cumulative days elapsed from the anchor epoch and maps to the corresponding Nepali year, month, and day.
- `bsToAd(bsYear, bsMonth, bsDay)`: Resolves exact Gregorian date.
- `getTodayBs()`: Emits human-readable formatted string (e.g. `Bhadra 26, 2083`).
- **Data Persistence**: Every entity in the system (Orders, Cheques, Store Audits, Dispatches) stores dual timestamp properties: `orderDateAd` and `orderDateBs`.

---

3. Role-Based Access Control & User Roles

- **Source Code**: `src/lib/auth.ts`, `src/components/common/PageGuard.tsx`

The application enforces role-based access at both the page route level (via) and the data filtering layer:

USER ROLE	DEFAULT LANDING PAGE	SCOPE OF PERMISSIONS & VISIBILITY
`SUPER_ADMIN`	`/dashboard`	Complete unconstrained read/write access to all brands, staff accounts, system settings, database migrations, and financial configurations.
`ADMIN`	`/dashboard`	Full operational management across orders, field teams, inventory dispatches, and review logs. Cannot delete brands or super admin users.
`BRAND_MANAGER`	`/dashboard` (Brand-Filtered)	Restricted strictly to their assigned brand portfolio (e.g. Forya only, or Pee Safe & FURR only). Sees filtered orders, customers, and revenue.
`SALES_OFFICER`	`/offline-sales/beat-plan`	Optimized for mobile view. Sees only their assigned beat stores, own call logs, own collected cheques, own shelf photos, and personal compliance scorecard.
`DATA_ENTRY`	`/orderflow/extractor`	Dedicated access to order intake, WhatsApp extractor, manual entry, delivery failure resolution, and returns processing.
`VIEWER`	`/dashboard` (Read-Only)	Read-only executive reporting access. All creation, editing, and status update actions are disabled.

---

4. Dynamic WhatsApp Extractor Engine (Fix 67 & 72)

- **Source Code**: `src/lib/whatsapp-extractor.ts`, `src/app/orderflow/extractor/page.tsx`

Core Operating Principle

The extractor never hardcodes product names, combos, prices, or keywords in source code. On every extraction execution, it pulls live catalog definitions from `db.getProducts()` and `db.getCombos()`.

Key Capabilities:

1. **Multi-Message Batch Ingestion**:

- Supports pasting individual orders or batches separated by `---` dividers or WhatsApp timestamp export headers:

`[12/09/26, 10:15:30 AM] Customer Name: ...`

1. **Customer & Delivery Info Extraction**:

- Matches Nepali mobile numbers (`+977`, `98XXXXXXXX`, `97XXXXXXXX`).
- Identifies address, landmarks, and delivery instructions.
- Maps locations against standard Nepal provinces and districts.

1. **Keyword Tag Matching & Nicknames**:

- Matches raw text against each product's `extractorKeywords` array.

- Recognizes abbreviations and transliterated terms (e.g., `rollon`, `peach roll on`, `serum`, `acne patch`, `ph lipstick`).

1. **Mixed Orders & Price-Based Combo Collapse**:

- Seamlessly extracts orders containing both combo bundles and standalone SKUs.
- **Price Collapse Rule**: When a combo has `priceDetectionEnabled: true` and the customer message states a total price matching that combo (e.g. `Rs. 999`), the engine automatically collapses ambiguous item text into that combo.

1. **Price Discrepancy Validation**:

- **Delta** `< 5\%`: Accepted automatically.
- **Delta** `5\% \text{–} 15\%`: Displays amber warning pill (`⚠️ Stated Rs. 1,000 vs calc Rs. 1,100 (9.1% diff)`).
- **Delta** `> 15\%`: Flagged for mandatory review (`🚨 Large Price Discrepancy`).
- **Stated Price** `>` **Calculated MRP**: Flagged (`⚠️ Stated price exceeds catalog MRP`).

1. **Live Catalog Sync Status Bar**:

- Displays real-time status: `🟢 Catalog Synced • 14 SKUs, 6 Combos loaded • Last updated just now`.

---

5. Daily Orders Live Cross-Check Engine (Fix 68)

- **Source Code**: `src/app/orderflow/daily-orders/page.tsx`

A real-time monitoring and cross-checking dashboard designed for office order coordinators to verify incoming WhatsApp orders against warehouse dispatch slips.

Visual & Event Feedback Architecture

- **Pub/Sub System**: Subscribes to `OrderRealtimeEvent` from `db-service.ts`.
- **Live Indicator**: Header displays `🟢 LIVE · 2 Viewers` with active listener indicators.
- **Green Slide-In Flash**: Triggers whenever an order is entered via Extractor or Manual Entry with a toast notification.
- **Amber Flash & Audit Trail**: Triggers when an order is updated, showing an audit trail tag (e.g., `Edited by Sita Sharma: Updated address`).
- **Fade-Out Animation**: Triggers when an order is cancelled or deleted.
- **Stated vs. Calculated Price Cross-Check**: Highlights price differences in dedicated comparison columns.
- **Segregated Marketing Sends**: Marketing and influencer seeding orders display with a distinct purple left border and `MARKETING` pill.
- **Sticky Metrics Footer**: Displays Total Orders, Commercial Revenue, Average Order Value (AOV), Delivery Failures, and Marketing Sends.

---

6. Smart Selling Assist Engine (Fix 46)

- **Source Code**: `src/lib/db/db-service.ts`, [`src/app/offline-sales/stores/[id]/page.tsx`](file:///Users/devanshagarwal/Documents/TEST/forya-admin/src/app/offline-sales/stores/[id]/page.tsx#L661)

Designed for field sales officers visiting retail pharmacies and grocery marts to maximize secondary order bookings using location-based intelligence.

The 4 Analytical Layers

A. 3km Neighborhood Bestsellers (`getAreaBestsellers`)

Uses the spherical **Haversine formula** to measure distances to all surrounding retail outlets:

$$d = 2 R \arcsin \left(\sqrt{\sin^2\left(\frac{\Delta \text{lat}}{2}\right) + \cos(\text{lat}_1)\cos(\text{lat}_2)\sin^2\left(\frac{\Delta \text{lon}}{2}\right)} \right)$$

where $R = 6,371\text{ km}$. It aggregates all secondary bookings within 3 km and ranks the Top 5 fastest-selling SKUs.

B. Distribution Whitespace Gaps (`getDistributionGaps`)

Cross-references the 3 km bestsellers against the current store's ordering history:

$$\text{Whitespace Gaps} = \{ \text{Neighborhood Bestsellers} \} \setminus \{ \text{SKUs Carried by Current Store} \}$$

Flags high-velocity products that neighboring competitors sell successfully but this store does not carry.

C. Shelf Depletion & Restock Forecasting (`getRestockRecommendations`)

Evaluates shelf audits over the previous 7-day audit cycle:

$$\text{Daily Depletion Rate} = \frac{\text{Previous Audit Stock} - \text{Current Shelf Stock}}{7\text{ days}}$$

$$\text{Days Until Empty} = \frac{\text{Current Shelf Stock}}{\text{Daily Depletion Rate}}$$

- **Urgent Warning**: Triggered if $\text{Days Until Empty} \le 3\text{ days}$.
- **Suggested Order**: Quantities calculated to cover a **20-day buffer**, rounded to box case sizes (6, 12, or 24 units).

D. 1-Tap Quick Booking

The `[⚡ Book 1-Tap Replenishment Order]` button automatically composes a balanced secondary order containing the top neighborhood bestsellers and critical restock items directly into the dispatch queue.

---

7. Customer Karma & 360 Intelligence (Fix 69)

- **Source Code**: `src/lib/karma-calculator.ts`, [`src/app/customers/[phone]/page.tsx`](file:///Users/devanshagarwal/Documents/TEST/forya-admin/src/app/customers/[phone]/page.tsx)

Customer Karma Score: 0 to 100 Scale

Starting baseline: **70 points** (or **85 points** upon placing first order).

$$\text{Updated Score} = \max\Big(0, \min\big(100, \text{Current Score} + \sum \Delta \text{Points}\big)\Big)$$

Karma Scoring Rules

EVENT TRIGGER	POINTS DELTA	BUSINESS IMPACT
Order Placed	**+2**	Customer initiates an order
Order Delivered	**+1**	Package successfully accepted
Prepaid Payment	**+1**	Zero COD risk payment
Review Submitted	**+3**	Engaged customer feedback
Positive Rating Bonus	**+2**	4 or 5 star rating
5 Clean Orders Milestone	**+10**	5 consecutive delivered orders with 0 returns/failures
Complaint Filed	**-2**	Service or packaging dissatisfaction
Actual Return	**-3**	Return of delivered goods
Delivery Failure	**-5**	Phone unanswered, unreachable at door
Order Cancellation	**-5**	Cancelled after packing/dispatch
COD Refusal at Doorstep	**-8**	Rider arrives but customer refuses to accept/pay

Automatic Tiers & Risk Flags

- `< 20`: `BLACKLISTED` — Advance payment mandatory before dispatch; COD strictly disabled.
- `20 – 39`: `AT_RISK` — High failure risk; requires telephone confirmation call before dispatch.
- `40 – 74`: `REGULAR` — Standard operational handling.
- `≥ 75` **(5+ orders)**: `LOYAL` — Preferred customer status.
- `≥ 85` **(10+ orders, 0 returns)**: `VIP` — White-glove handling, free gift samples, priority dispatch.
- **Behavioral Flags**:
- `Returns ≥ 3` \rightarrow `Serial Returner`
- `Delivery Failures ≥ 2` \rightarrow `Non-Receiver`
- `Cancellations ≥ 3` \rightarrow `Order Canceller`

8. Offline Field Sales & Beat Operations (Fix 44, 45, 48, 49)

- **Source Code**: `src/app/offline-sales/beat-plan/page.tsx`, `src/lib/route-optimizer.ts`

Sales Officer Daily Workflow

1. **Morning Shift Start**: Sales officer toggles shift status (🟢 Online / ⚪ Offline). Pinned morning directives from management display at the top.
2. **Smart Day Planner (Route Optimization)**:

- Groups stores by priority: ★★★ **High Priority** (must visit today), ★★ **Medium Priority**, ★ **Routine**.
- Solves the Traveling Salesperson Problem (TSP) using GPS coordinates to order visits by nearest-neighbor distance.

1. **GPS Check-In**:

- Validates that the officer is within **100 meters** of the store's recorded latitude/longitude.
- If outside 100m, prompts for reason (e.g. "GPS drift", "Meeting manager at cafe") and logs an override flag for admin review.

1. **In-Store Execution Hub**:

- **Physical Stock Count**: Audits shelf inventory against safety threshold levels.
- **Secondary Booking**: Books new orders with Smart Selling Assist recommendations.
- **Before/After Shelf Photos**: Captures planogram compliance and counter display placement.
- **Cheque Collection**: Records cheque number, issuing bank, due date, amount, and photo of physical cheque.

1. **GPS Check-Out**: Records total visit duration and moves to the next route stop.
2. **End-of-Day Scorecard**:

- Calculates beat compliance percentage, total calls made, cheques collected, and secondary revenue.
- Automatically dispatches a daily summary push notification to admins.

---

9. Store Health Score Engine (Fix 34)

- **Source Code**: `src/lib/db/db-service.ts`

Evaluates the commercial relationship and financial health of distributor branches and retail accounts on a **0 to 100 scale**:

```
\text{Store Health Score} = S_{\text{Regularity}} + S_{\text{Payment}} + S_{\text{Stock}}
+ S_{\text{Cooperation}} + S_{\text{Issues}}
```

The 5 Weighted Factors

1. **Order Regularity (25 pts)**: Frequency of reorders (Monthly = 25 pts, Bi-monthly = 15 pts, Irregular = 0 pts).

2. **Payment Discipline (25 pts)**: Percentage of cheques cashed on time (100% on time = 25 pts; deductions for bounced cheques).
3. **Stock Maintenance (20 pts)**: Consistency of shelf stock levels remaining above minimum threshold.
4. **Visit Cooperation (15 pts)**: Percentage of scheduled beat visits where the sales officer was permitted to conduct audits.
5. **Issue Rate (15 pts)**: Ratio of returns/complaints to total orders (`0\text{–}2\% = 15\text{ pts}`, `>10\% = 0\text{ pts}`).

---

10. Marketing & Influencer Seeding Pipeline (Fix 60 & 71)

- **Source Code**: `src/app/orderflow/manual-entry/page.tsx`, `src/app/marketing/influencers/page.tsx`

Complete Commercial Segregation

1. **Zero Revenue Impact**:

- All PR packages and influencer product seeding orders have `totalAmount: 0`.
- Excluded from commercial delivery failure rates, return rates, average order value, and gross revenue metrics.

1. **Influencer Directory Search & Preview**:

- Search by creator name, social handle, or niche category.
- Preview card displays star rating (`⭐⭐⭐ Great`), platform (`Instagram`), follower count, total sends count, and content deliverables received.

1. **Delivery Defaulting**:

- Checking "Marketing / Influencer Send" automatically defaults delivery method to `Pick-up` (or Direct Send).

1. **Auto-Increment Counter**:

- Order submission automatically calls `db.incrementInfluencerSends(influencerId)`.

---

11. Product, Combo & Pricing Catalog Management (Fix 70b & 72)

- **Source Code**: `src/app/products/page.tsx`

Catalog Architecture

- **Odoo References**: Completely eradicated project-wide.
- **Pricing Hygiene**: Internal financial pricing fields (`wholesalePrice`, `dtpPrice`, `costPrice`) are eliminated from product creation forms. Only customer-facing catalog `mrp` and `unitPrice: mrp` are managed.
- **Extractor Keyword Tag Editor**:

- Interactive keyword pills for every product and combo.
- `[🔄 Auto-Suggest]` **Engine**: Generates brand-less names, individual product tokens, and transliterated terms (e.g. `roll on`, `rollon`, `peach roll on`, `serum`, `nose strips`).
- **Promotional Combo Builder**:
- Bundles 1 to 3 component SKUs with quantities.
- Dynamically calculates total individual MRP, bundle combo price, and consumer savings percentage.
- **Price-Based Collapse Toggle**: Enables WhatsApp extractor to match the combo whenever total price is stated.
- **Audit Trail**: Records date, modifier, and price change history.

---

12. Order Delivery Failures vs. True Returns (Fix 22 & 23)

- **Source Code**: `src/app/orderflow/delivery-failures/page.tsx`, `src/app/reviews/returns/page.tsx`

The platform strictly separates **logistical delivery attempts** from **product returns**:

Delivery Failures (Logistics Non-Delivery)

- **Definition**: The customer never received the package (e.g. "Call Not Received", "Customer Out of Valley", "Incomplete Address", "Customer Refused COD").
- **Workflow**: Logged under `/orderflow/delivery-failures`. Supports scheduling re-attempts (Attempt 1, 2, or 3), rider reassignment, and phone follow-up logging. Deducts 5 karma points.

Actual Returns (Delivered Product Issues)

- **Definition**: The package was delivered, but the customer returned the product due to product/order quality issues.
- **Categories**: Leaking Product, Damaged Packaging, Wrong Product Sent, Expired Product, Product Reaction/Allergy.
- **Workflow**: Logged under `/reviews/returns`. Tracks return inspection status, customer replacement dispatch, or refund issuance.

---

13. Operations Core: Tasks, Calendar & Audit Activity (Fix 37, 38, 66)

- **Source Code**: `src/app/opscore/tasks/page.tsx`, `src/app/opscore/calendar-tasks/page.tsx`, `src/app/opscore/activity/page.tsx`

1. **Predefined Task Templates (Fix 37)**:

- Allows 1-click creation of multi-step task workflows:
- **New Store Onboarding** (5 steps: Introduce portfolio `\rightarrow` Photograph shelf `\rightarrow` First stock count `\rightarrow` Collect first order `\rightarrow` Upload agreement).
- **Cheque Recovery Sequence** (3 steps: Call vendor `\rightarrow` Visit store `\rightarrow` Escalate to Admin).

- **Complaint Resolution** (4 steps: Contact customer \rightarrow Investigate batch \rightarrow Provide replacement \rightarrow 3-day follow-up).
- **Monthly Stock Audit** (3 steps: Complete stock count \rightarrow Photograph shelf \rightarrow Discrepancy report).

1. **Bikram Sambat Visual Calendar (Fix 66)**:

- Day, Week, and Month matrix plotting tasks, scheduled store beats, and public holidays side-by-side.

1. **Admin Field Push Summaries (Fix 38)**:

- When a sales officer taps "End My Day", an automated summary push notification is dispatched to all Admins summarizing stores visited, compliance percentage, calls made, cheques collected, and field time.

---

14. Automated Cron Services & Holiday Calendar (Fix 32 & 33)

- **Source Code**: `src/app/api/cron/daily-alerts/route.ts`, `src/app/api/cron/weekly-report/route.ts`

1. **Daily Morning Alerts (8:00 AM NPT)**:

- Evaluates system state for critical conditions:
- 🔴 **Red Alerts**: Bounced cheques, customer karma dropped < 20, urgent unassigned complaints, GPS check-in overrides.
- 🟡 **Amber Alerts**: Cheques overdue by 1 day, stores unvisited for >10 days, delivery failure rate >15\%, low distributor stock.

1. **Friday Operations Email Report (8:00 PM NPT)**:

- Computes 7-day rolling performance: total orders by brand, delivery failure rates, actual returns breakdown, beat compliance per sales officer, and cash flow summary.

1. **Holiday Calendar Engine (Fix 33a)**:

- Pre-loaded with Nepali public holidays (Dashain, Tihar, Chhath, Teej, Holi, etc.).
- Adjusts field compliance expectations (e.g. a 5-day working week does not penalize officers for festival days).

---

15. Excel Data Import & Master Google Sheets Export (Fix 39, 42, 43)

- **Source Code**: `src/app/settings/components/DataImport.tsx`, `src/app/opscore/export/page.tsx`

Data Import Engine (Fix 39)

Seamlessly ingests legacy Excel spreadsheets across 4 standardized formats:

1. **Orders Import**: Reads monthly tabs from `FORYA_MasterOrderEntry` format.

2. **Returns Import**: Reads per-brand tabs from `BB_RETURN_SHEET` format and separates true returns from delivery failures.
3. **Offline Sales Import**: Reads stores, branches, stock counts, and dispatches from `PEESAFE_OFFLINE_SALES` format.
4. **Daily Order Counts**: Ingests historical channel tallies from `Forya_SOCIAL_ORDER_COUNT`.

Google Sheets Aligned Export (Fix 42 & 43)

Exports data in the exact column layouts expected by corporate management:

- **Orders Export**: `S.N. | Order ID | Order Type | Order Date (BS) | Order Date (AD) | Platform | Customer Name | Phone Number | Province | District | Brand | Product Category | Product Name | Qty | Delivery Status | Remarks`
- **Daily Social Order Counts Table**: Auto-computes daily date rows with columns for Website, Instagram, Facebook, WhatsApp, TikTok, Total, and individual product volumes.

---

16. Production Database Schema & Performance Indexing (Fix 59)

- **Source Code**: `supabase/schema.sql`

The production Postgres database is configured with **16 specialized performance indexes** to ensure sub-10ms query times at scale:

1. `idx_orders_brand_date` ON `orders(brand_id, order_date_ad DESC)`
2. `idx_orders_customer_phone` ON `orders(customer_phone)`
3. `idx_orders_delivery_status` ON `orders(delivery_status)`
4. `idx_orders_platform` ON `orders(platform)`
5. `idx_delivery_failures_date` ON `delivery_failures(date_ad DESC)`
6. `idx_delivery_failures_reason` ON `delivery_failures(reason)`
7. `idx_returns_brand_date` ON `returns(brand_id, date_ad DESC)`
8. `idx_feedback_brand_sentiment` ON `feedback_entries(brand_id, sentiment)`
9. `idx_cheques_status_cash_date` ON `cheques(status, cash_date_ad)`
0. `idx_cheques_store` ON `cheques(store_id)`
1. `idx_beat_officer_date` ON `beat_plans(officer_id, date_ad)`
2. `idx_beat_status` ON `beat_plans(status)`
3. `idx_stock_counts_entity_date` ON `stock_counts(entity_id, date_ad DESC)`
4. `idx_customers_phone` ON `customers(phone)`
5. `idx_customers_karma` ON `customers(karma_score)`
6. `idx_activity_trail_entity` ON `activity_logs(entity_type, entity_id)`

---

17. Complete Route Map & File Directory Index

```
src/
├── app/
│   ├── layout.tsx                          # Global Root Layout (Fonts, Metadata, PWA me
│   ├── page.tsx                            # Root Redirect (Role-based auth redirect)
│   ├── login/page.tsx                      # Authentication Screen (Phone/Password)
│   ├── dashboard/page.tsx                  # Executive Command Dashboard (Morning Brief,
│   ├── orderflow/
│   │   ├── extractor/page.tsx              # Dynamic WhatsApp Extractor (Catalog-Driven)
│   │   ├── daily-orders/page.tsx           # Daily Orders Live Cross-Check (Pub/Sub Flas
│   │   ├── manual-entry/page.tsx           # Manual Order Entry (Drafts, Influencers)
│   │   └── delivery-failures/page.tsx      # Delivery Failures Logging & Re-Attempts
│   ├── offline-sales/
│   │   ├── beat-plan/page.tsx              # Sales Officer "My Day" Beat Hub
│   │   ├── stores/page.tsx                 # Store Directory & Health Badges
│   │   ├── stores/[id]/page.tsx            # Store 360 Intel & Smart Selling Assist
│   │   ├── calls/page.tsx                  # Field & Telephonic Call Logger
│   │   ├── cheques/page.tsx                # Cheque Tracking & Custody Handover
│   │   ├── dispatch/page.tsx               # Warehouse Dispatches & Packing Slips
│   │   ├── documents/page.tsx              # Field Document Vault (Invoices, Agreements)
│   │   ├── shelf-photos/page.tsx           # Merchandising Before/After Photos
│   │   ├── live-tracking/page.tsx          # Live GPS Field Officer Map
│   │   └── scorecard/page.tsx              # Officer Compliance & Revenue Scorecard
│   ├── customers/
│   │   ├── page.tsx                        # Customer 360 Directory (Brand Filter Pills)
│   │   ├── [phone]/page.tsx                # Customer Profile (Brand Engagement & Timeli
│   │   └── insights/page.tsx               # Retention Analytics (Cohorts, Churn, Funnel
│   ├── products/
│   │   ├── page.tsx                        # SKU & Combo Master (Extractor Keywords)
│   │   └── [id]/page.tsx                   # Product 360 Intel & Seasonal Trend Sparklin
│   ├── reviews/
│   │   ├── feedback/page.tsx               # Customer Reviews & Photo Evidence
│   │   ├── returns/page.tsx                # Actual Returns Processing
│   │   └── categories/page.tsx             # Dynamic Checkbox Categories & Tags
│   ├── marketing/
│   │   ├── influencers/page.tsx            # Influencer Directory & Ratings
│   │   ├── influencers/[id]/page.tsx       # Influencer Portfolio & Deliverables
│   │   └── sends/page.tsx                  # PR Seeding Send Records
│   ├── opscore/
│   │   ├── tasks/page.tsx                  # Task Manager & Templates
│   │   ├── calendar-tasks/page.tsx         # Bikram Sambat Visual Calendar
│   │   ├── activity/page.tsx               # Audit Trail & EOD Officer Logs
│   │   └── export/page.tsx                 # Excel & Google Sheets Aligned Export
│   ├── settings/page.tsx                   # System Settings, Holidays & Excel Import
│   └── shared/[token]/page.tsx             # Tokenized External Read-Only Reports
└── lib/
    ├── db/
    │   ├── db-service.ts                   # Unified Data Layer & Real-Time Pub/Sub
    │   ├── mock-data.ts                    # Seeded Catalog, Beats, Stores & Orders
    │   ├── supabase-client.ts              # Supabase Browser Client
    │   └── supabase-server.ts              # Supabase Server Client
    ├── nepali-calendar.ts                  # Bikram Sambat (BS) ↔ Gregorian (AD) Engine
    ├── karma-calculator.ts                 # Customer Karma Scoring & Segmentation
    ├── whatsapp-extractor.ts               # Catalog-Driven Order Extractor Engine
    ├── route-optimizer.ts                  # Traveling Salesperson Beat Optimizer
    ├── auth.ts                             # Authentication & Session Handlers
```

```
    ├── validation.ts                          # Nepali Phone & Field Validators
    └── types.ts                               # Full TypeScript Contracts & Interfaces
```

Manual compiled and verified for Brand Bucket Command Center 2.0.